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Proposal Review 2 : 2502380

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Agency Name:	National Science Foundation
Agency Tracking Number:	2502380
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Tropicalizations of Shimura Varieties
Rating:	Multiple Rating: (Excellent/Very Good)

Review

Summary

Proposal Synopsis

The broad goal of the project is to study Shimura varieties and in particular to understand their cohomology. To achieve this goal the PI proposes to develop the theory of tropical(ization of Shimura varieties and use the combinatorics of that space to compute cohomology.

Intellectual Merit

Strengths

Rating: V

This is a very well-written proposal. The PI has a great record of past research accomplishments. The proposal outlines ways each of the proposed problems will be addressed. First prop project: The research program is clearly outlined after stating the research problem (Problem 2.6). This is work in progress and seems the PI has made significant progress already in orde make this conjecture. Second suggested project: this seems to be a generalization of previous results of the PI. The third part of the project: is computational, based on adapting code the co-authors used for previous work.

Weaknesses

The projects proposed seem to be work in progress or generalizations of past work.

Broader Impacts

Strengths

Rating: E

The PI has an impressive list of events organized for someone this early in their career. The PI has immediate plans for conference organization for the upcoming year(s). The PI also has

to support mathematicians from historically underrepresented backgrounds through work with SLMath and Dartmouth.

Weaknesses

The ideas for the REU and mentorship programs as well as the SLMath workshop are not very fleshed out. The PI is already in the process of mentoring students.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

Rating: E

The PI has held NSF Graduate Research Fellowship and NSF Postdoctoral Research Fellowship: The results are total of 13 papers (some published in very prestigious journals) and huge involvement in organizing and mentoring activities.

Summary Statement

this is a good proposal with a strong chance of success

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